Lab 1: Introduction to Packet Sniffing and Wireshark

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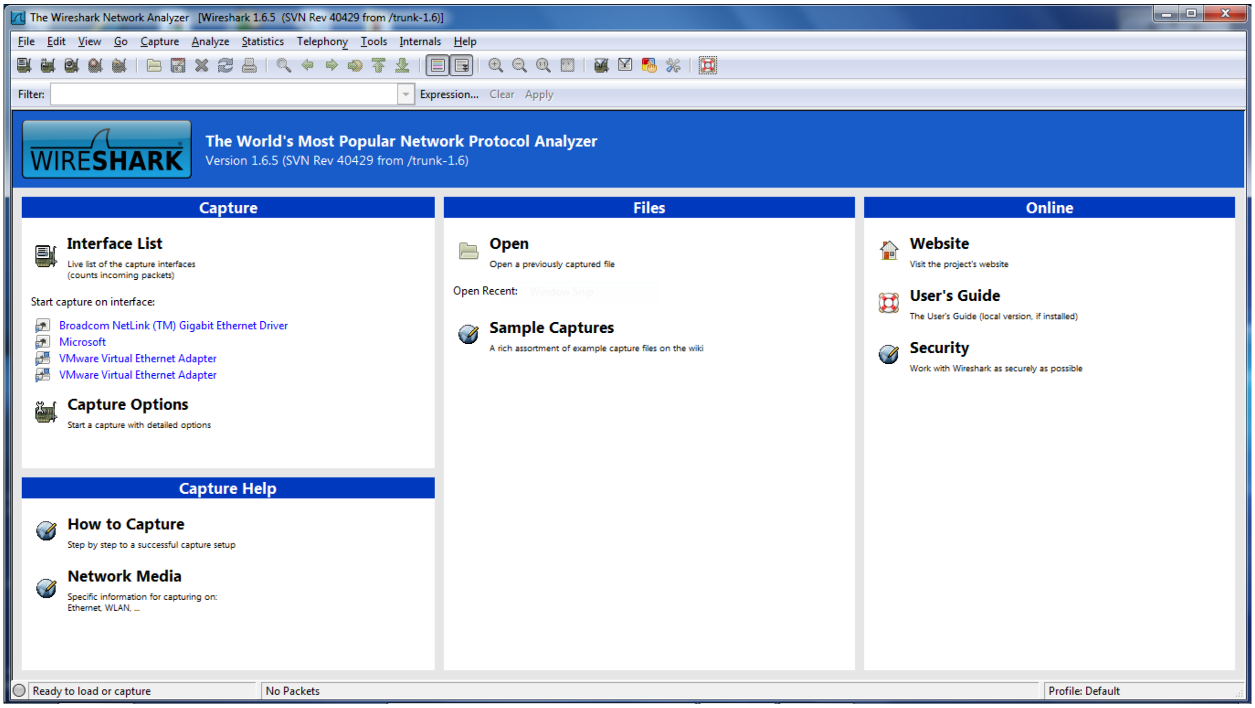
Part A: Introduction

1.Question: What information is available by following each of the start-up screen links?



Under the capture section, the interface list shows the available interfaces that you will be able to capture packets from. The capture options will change your configuration of how you want to capture the packets. Under capture help, there is information of how to capture from an interface and network media for different types of devices. Under the files section, you can open previously captured files and open sample captures to play with the application. Under the online section, there are additional resources such as visiting the website, opening the user's guide, and working with Wireshark security. This is an older version Wireshark, yet the updated version has similar features.

2.Question: What information is displayed in the packet list pane, the packet details pane, the packet bytes pane and the status bar?

Packet list pane shows a summary of all captured packets including packet number, time, source, destination, protocol, length, and additional info. The packet details pane shows details of the structure and data of a single packet shown as tabs of information for each protocol layer. The packet bytes pane shows binary data as hexadecimal values of the selected packet. The status bar displays the total number of packets, the number of packets displayed, the number of packets dropped, and the current profile (configuration) selected.

3. Question: Examine the packet details pane. In relation to the OSI model, what order is the information presented?

The packet details pane lists the layers from data link, network, transport, and application ordered from top to bottom. For an example packet received using the TCP protocol, the following tabs of the pane include Frame #, Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Data.

4. Question

Expand the fields in the packet details pane to answer the following question.

1. What is the source MAC address of the frame?
   00:00:c0:9f:a0:97

2. What is the destination MAC address of the frame?
   00:a0:cc:3b:bf:fa

3. What is the source IP address of the packet?
   192.168.0.1

4. What is the destination IP address of the packet?
   192.168.0.2

5. What is the source port of the segment?
   23

6. What is the destination port of the segment?
   1550

7. What is the data payload?
   PING www.yahoo.com (204.71.200.67)

8. What company manufactured the source NIC? How does Wireshark know this?
   The company that manufactured the source network interface card is called Western Digit. Wireshark knows this because it uses the first three bytes of the MAC address, also known as the Organizationally Unique Identifier. Wireshark has a table that maps these OUI values to the manufacturers.

5. Question: Expand the fields in the packet details pane to answer the following questions.

1. What is the source MAC address of the frame?
   00:00:c0:9f:a0:97

2. What is the destination MAC address of the frame?
   00:a0:cc:3b:bf:fa

3. What is the source IP address of the packet?
   192.168.0.1

4. What is the destination IP address of the packet?
   192.168.0.2

5. What is the source port of the segment?
   23

6. What is the destination port of the segment?
   1550

7. What is the data payload?
   64 bytes from 204.71.200.76

8. Based on the contents of frame 55 and frame 57, what even has taken place? Be as detailed as possible.
   Frame 55 pinged yahoo.com, the website returned the ping with 64 bytes of data. The content return of a ping varies, but it may include a timestamp that shows the time of transmission or other statistics of the packet.

7.Question: Based on the content of these frames, what event(s)has/have taken place during the exchange of these frames? Hint: there is no payload and observe the SYN and ACK bit values.

The event that has taken place within the first three frames is the TCP handshake. The TCP handshake involves a three-way handshake to establish a reliable connection. SYN stands for synchronize and ACK stands for acknowledge, the client sends a SYN to the server which returns an SYN/ACK back to the client, then the client sends an ACK back to the server.

8.Question: Based on the content of these frames what event has taken place? Hint: Pay attention to the FIN and ACK flags.

Frames 89 to 92 describes a four-way handshake to close the TCP connection. The client sent a FIN (finish) packet to the server to tell it that it is done sending data. The server acknowledges the FIN of the client and sends an ACK packet back. The server sends its own FIN packet to the client when the server is finished sending its data. Lastly, the client sends an ACK packet to acknowledge the FIN packet of the server. After the last pack, the TCP connection is closed.

9.Question: The captured file that you are working with is a single Telnet session. To follow the session select one of the packets, right-click and then select Follow-> TCP Stream. Copy the output of the stream and include it in your report for this lab.

```
.......... ..!..".'.....#
..%
..%
.......... ..!..".."....
....P. ....".....b........b.... B.
......................."....
..'.....#..&..&..$
..&..&..$
.. .....#.....'.........
.. .9600,9600....#.bam.zing.org:0.0....'..DISPLAY.bam.zing.org:0.0......xterm-color..
...
...
.....!......
......
.."...........
OpenBSD/i386 (oof) (ttyp2)

login:
fake

...
...
Password:
user


...
...
Last login: Sat Nov 27 20:11:43 on ttyp2 from bam.zing.org
Warning: no Kerberos tickets issued.
OpenBSD 2.6-beta (OOF) #4: Tue Oct 12 20:42:32 CDT 1999

Welcome to OpenBSD: The proactively secure Unix-like operating system.
```

Please use the sendbug(1) utility to report bugs in the system.
Before reporting a bug, please try to reproduce it with the latest
version of the code. With bug reports, please try to ensure that
enough information to reproduce the problem is enclosed, and if a
known fix for it exists, include that as well.

```
$
/sbin/ping www.yahoo.com

PING www.yahoo.com (204.71.200.67): 56 data bytes
64 bytes from 204.71.200.67: icmp_seq=0 ttl=241 time=69.885 ms
64 bytes from 204.71.200.67: icmp_seq=1 ttl=241 time=73.591 ms
64 bytes from 204.71.200.67: icmp_seq=2 ttl=241 time=72.302 ms
64 bytes from 204.71.200.67: icmp_seq=3 ttl=241 time=73.493 ms
64 bytes from 204.71.200.67: icmp_seq=4 ttl=241 time=75.068 ms
64 bytes from 204.71.200.67: icmp_seq=5 ttl=241 time=70.239 ms

.....
.....
.--- www.yahoo.com ping statistics ---
6 packets transmitted, 6 packets received, 0% packet loss
round-trip min/avg/max = 69.885/72.429/75.068 ms
$
ls

$
ls -a

. .. .cshrc .login .mailrc .profile .rhosts
$
exit
```

10.Question:

Use Wireshark and any available Internet resources to answer the following questions?

1. Frame 53 is labeled as a retransmission. Why did this occur?
   The retransmission occurred because the client did not acknowledge a segment of the data in time, so the sender retransmits that segment to ensure that the client fully received the data.

2. Frame 72 is labeled as a Malformed Packet. What does this mean?
   A malformed packet is a packet that does not follow the protocol rules of the network or does not have the expected format.

3. Frame 78 has a PSH flag. What does this mean?
   The PSH flag stands for push and tells the receiving device to send the data to the receiving application right away.

4. Understanding that the capture file you are using is of a single Telnet session, outline the process that TCP/IP uses to establish, maintain and terminate a Telnet session.
   TCP uses a three-way handshake with SYN and ACK packets, uses ACK packets to ensure reliable data transfer, and a four-way handshake using FIN and ACK packets to close the connection.

Part B: Performing a Simple Data Capture

11.Question: *Prepare a summary of the types of traffic that Wireshark detects on the interface.*

Wireshark detects all types of network traffic. Some of the protocols captured on my Wi-Fi are AJP13, ARP, BJNP, MDNS, QUIC, SSDP, TCP, and TLSv1.2.

12.Question:Answer the following questions based on your capture.

1. What are the source and destination MAC addresses for the ARP request?
   Source: SagemcomBroa_1d:3a:28 (0c:ac:8a:1d:3a:28)
   Destination: Intel_e3:99:1e (60:dd:8e:e3:99:1e)

2. What are the source and destination IP addresses for the ARP request?
   Source: 192.168.2.1
   Destination: 192.168.2.19

3. What data is contained in the ARP request?
   ARP requests have the MAC and IP addresses of the destination and sender, the protocol being used., and information about the hardware.

4. What are the source and destination MAC addresses for the ARP reply?
   Source: Intel_e3:99:1e (60:dd:8e:e3:99:1e)
   Source: SagemcomBroa_1d:3a:28 (0c:ac:8a:1d:3a:28)

5. What are the source and destination IP addresses for the ARP reply?
   Source: 192.168.2.19
   Destination: 192.168.2.1

6. What data is contained in the ARP reply?
   The data contained in the ARP reply is the same as the data contain in the ARP request.

13.Question: Select File | Save As and save your capture file using your student number as the file name